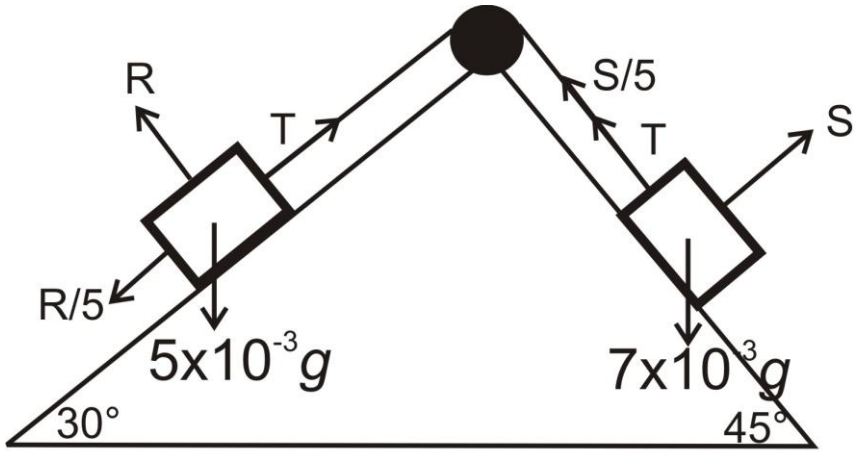
CAP exam 2010: Solutions:

Multiple Choice:

1: b	7: b	13: b	19: a
2: d	8: a	14: e	20: d
3: a	9: a	15: b	21: c
4: d	10: c	16: b	22: d
5: a	11: b	17: a	
6: a	12: d	18: c	

Long Answer:

1. Let T be the tension in the string, and R and S bet the reactions between the masses and their respective planes.



Then

$$R = 5\times10^{-3} g\cos 30^\circ$$
$$S = 7\times10^{-3} g\cos 45^\circ$$

g = 9.8 m/s^2 and the frictional forces are $\frac{R}{5}$ and $\frac{S}{5}$.

Let f be the acceleration of the 7 gram mass down its plane and of the 5 gram mass up its plane. The equations of motion of the masses are:

$$7\times10^{-3} g\sin 45^\circ - \frac{1}{5}7\times10^{-3} g\cos 45^\circ - T = 7\times10^{-3} f$$
$$T - 5\times10^{-3} g\sin 30^\circ - 1\times10^{-3} g\cos 30^\circ = 5\times10^{-3} f$$

or

$$\frac{7\times10^{-3} g}{\sqrt{2}} - \frac{1}{5\sqrt{2}}7\times10^{-3} g - T = 7\times10^{-3} f$$
$$T - \frac{5\times10^{-3} g}{2} - \frac{\sqrt{3}\times10^{-3} g}{2} = 3.27\times10^{-3} f$$

Adding these together gives

$$\frac{g}{120}\left(28\sqrt{2}-25-5\sqrt{3}\right)=f$$
$$f=0.049g=0.481m/s^2$$

Thus the acceleration of the blocks is to the right (ie. the 7 gm mass moves downwards) at 0.481 m/s^2. From this we can calculate T, which is $T=0.0032N$.

The forces exerted by the 5 gm mass on the wedge are $5\times10^{-3}g\cos30^\circ$ perpendicular to the plane, and $1\times10^{-3}g\cos30^\circ$ up the plane. The component vertically downwards of these forces is

$$5\times10^{-3}g\cos^2 30^\circ-1\times10^{-3}g\cos30^\circ\sin30^\circ=(\frac{15-\sqrt{3}}{4})\times10^{-3}g$$

Similarly, the component vertically downwards of the forces exerted by the 7 gram mass on the wedge are

$$7\times10^{-3}g\cos^2 45^\circ-\frac{7}{5}\times10^{-3}g\cos45^\circ\sin45^\circ=\frac{21}{5}\times10^{-3}g$$

The downward component of the force exerted by the string on the pulley is

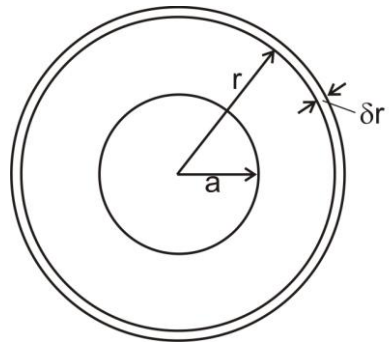
$$T\cos60^\circ+T\cos45^\circ=4.36\times10^{-3}g.$$

Adding up all the downward forces, we find

$$10\times10^{-3}g+4.36\times10^{-3}g+\frac{21}{5}\times10^{-3}g+\frac{15-\sqrt{3}}{4}\times10^{-3}g=21.88\times10^{-3}g=0.214N$$

Or the equivalent of weight of 21.88 gram (slightly less than the mass of the three blocks if they were not moving).

2. Consider a sphere surrounded by a spherical shell as shown in the figure. The thickness of the shell is δr. The current I flows perpendicular to the shell.



The resistance dR of the shell is

$$\delta R = \rho \frac{\text{length parallel to the current flow}}{\text{area perpendicular to the current flow}}$$

The resistance from *a* to *2a* is

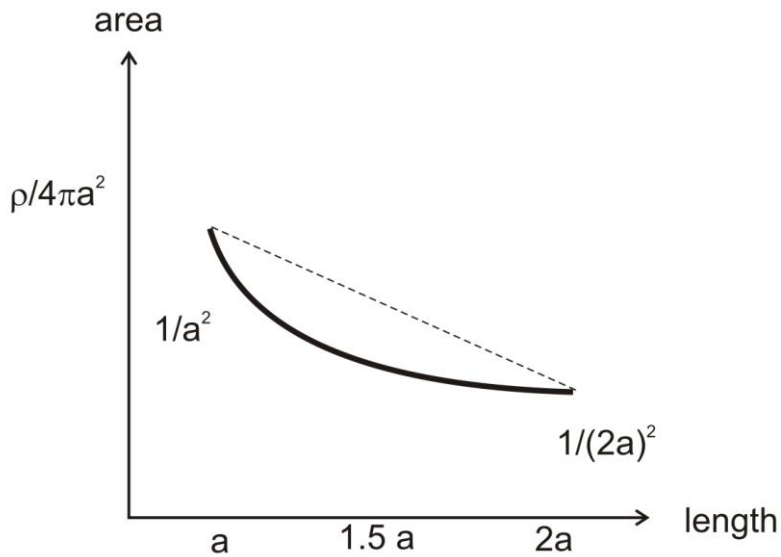
$$R\Big|_a^{2a} = \rho \int_a^{2a} \frac{dr}{4\pi r^2} = \frac{\rho}{4\pi}\left[\frac{1}{a} - \frac{1}{2a}\right] = \frac{\rho}{8\pi a}$$

$$R\Big|_a^{\infty} = \rho \int_a^{\infty} \frac{dr}{4\pi r^2} = \frac{\rho}{4\pi a} = 2 \times R\Big|_a^{2a}$$

For two spheres,

$$R_{Total} = 2 \times R\Big|_a^{\infty} = \frac{\rho}{2\pi a}$$

Graphically, this can be done in a number of ways. For instance, here is a typical way this can be approached:



One could represent the integral as a trapezium and get for the area under the curve (i.e. the approximation for the resistance between a and 2a)

$$\frac{1}{2}\left[\frac{1}{a^2}+\frac{1}{(2a)^2}\right]=\frac{5}{8a}\times\frac{\rho}{4\pi}$$

which is a little larger than the integral value. For the value between a and infinity, students can obtain a table of partial sums by estimating the area under the curve for different successive intervals of a, and demonstrate that the solution converges to a finite value. Please note that the exact sum of the inverse square series was solved by Euler and is quite beyond the brilliant highschool student (we think!). The subtleties of this result are one of the great advertisements for integral calculus.

b) At the surface of the sphere, the current is

$$I = 4\pi a^2 J_r$$

where J_r is the radial component of the current density. Thus

$$J_r = \frac{I}{4\pi a^2}$$

From Ohm's law, the radial component of the electric field is then

$$E_r = \frac{\rho I}{4\pi a^2}$$

If Q is the charge on the sphere, then the electric field is given by

$$E_r = \frac{1}{4\pi\varepsilon_o}\frac{Q}{a^2}$$

By comparison, then

$$Q = \rho I \varepsilon_0$$

How big is Q? If I is 1 A, ρ is of the order of 1 Ohm, and ε_0 is roughly 10^{-11} F/m, so Q is of order 10^{-11} c.

3. a) This comes down to showing that $N_0 e^{-\lambda T_{1/2}} = \frac{N_0}{2}$.

$$e^{-\lambda T_{1/2}} = \frac{1}{2}$$

$$\lambda T_{1/2} = \ln 2$$

$$T_{1/2} = \frac{\ln 2}{\lambda}$$

b) In the first reaction, the potassium nucleus captures an electron from the inner shell (since the electron has a finite probability of being inside the nucleus), and mutates into Argon. In the second reaction, a positron is emitted by the potassium radioactive nucleus (by standard beta+ decay), and the potassium mutates into Argon.

c) The reaction most likely to occur is that which requires the least change in total energy (or equivalently, total mass). Therefore, let us look at the energy balances. The total energy of the reaction is determined by the total mass of the particles before the decay minus that of the particles after the decay:

Electron capture:

$$Q_{ec} = (m_{potassium} + m_{electron} - m_{\text{argon}} - \nu)c^2$$

Positron emission:

$$Q_{pe} = (m_{potassium} - m_{electron} - m_{\text{argon}} - \nu)c^2$$

Subtracting the two we find that $Q_{ec} = Q_{pe} + 2m_{electron}c^2$. Therefore electron capture is less energetically demanding and is more likely to occur.

d) ${}^{40}_{19}\text{K} \rightarrow {}^{40}_{20}\text{C}a + {}^{0}_{-1}\text{e} +$ an antineutrino

which is a standard β^- decay

e) Rearranging this equation to solve for $\frac{K_f}{Ar_f}$:

$$\frac{t \ln 2}{T_{1/2}} = \ln(1 + \frac{Ar_f}{.109 K_f})$$

$$0.109(e^{\frac{t \ln 2}{T_{1/2}}} - 1) = \frac{Ar_f}{K_f}$$

Plugging in the given values for the half-life and the time since the formation of the rock gives $\frac{K_f}{Ar_f} = 0.003069$.